Gradient-Based Hull Shape Optimization for Wave Energy Converters via Hybrid Analytical–Finite-Difference Sensitivities

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Abstract

We present a gradient-based framework for optimizing the hull geometry of a wave energy converter (WEC) to maximize electrical power output. The approach exploits a bilevel structure: an inner nonlinear program (NLP) solved by IPOPT determines the optimal power take-off (PTO) control for a given hull shape, while an outer optimizer adjusts the hull shape parameters. The key contribution is a *hybrid gradient* that combines (i) an *analytical* Fiacco post-optimality sensitivity for the derivative of optimal power with respect to boundary element method (BEM) hydrodynamic coefficients, with (ii) *finite differences* through the BEM solver (Capytaine) for the derivative of hydrodynamic coefficients with respect to hull shape parameters. The chain rule links these two components, yielding an accurate gradient at a fraction of the cost of full finite-difference optimization. We demonstrate the method on the AquaHarmonics device, optimizing the main cylinder radius and draft under a regular wave condition.

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1 Introduction

The design of wave energy converters involves coupled decisions about hull geometry and control strategy. Hull shape determines the hydrodynamic response—added mass, radiation damping, excitation forces—which in turn governs the achievable power under optimal control. Traditional shape optimization relies on derivative-free methods (e.g., genetic algorithms, Nelder–Mead) because the shape-to-power pipeline involves non-differentiable components: mesh generation, BEM solvers, and NLP solvers.

We propose a hybrid approach that renders this pipeline *partially differentiable*. The expensive NLP solve is handled analytically via the Fiacco post-optimality sensitivity theorem (also known as the envelope theorem), while the BEM solver is treated as a black box and differentiated via finite differences. This reduces the cost of each gradient evaluation from $\mathcal{O}(n_{\text{shape}})$ full NLP solves to $\mathcal{O}(n_{\text{shape}})$ BEM solves plus a single NLP solve, often an order-of-magnitude speedup.

2 WEC Dynamics and Pseudo-Spectral Formulation

2.1 Equation of Motion

The dynamics of a floating body in waves are governed by the Cummins equation. In the frequency domain, for a single degree of freedom (heave), the equation of motion is

$$-\omega^2\big[M + A(\omega)\big]\hat{X}(\omega) + j\omega\big[B(\omega) + B_f\big]\hat{X}(\omega) + K\,\hat{X}(\omega) = \hat{F}_{\text{exc}}(\omega) + \hat{F}_{\text{PTO}}(\omega), \tag{1}$$

where:

- M is the rigid-body inertia matrix,
- $A(\omega)$ is the frequency-dependent added mass,
- $B(\omega)$ is the radiation damping,
- B_f is linear friction (supplementary damping),
- K is the hydrostatic stiffness,
- $\hat{F}_{\text{exc}}(\omega) = \hat{F}_{\text{FK}}(\omega) + \hat{F}_{\text{diff}}(\omega)$ is the wave excitation force (Froude–Krylov plus diffraction), and
- $\hat{F}_{\text{PTO}}(\omega)$ is the PTO force (a decision variable in the control optimization).

2.2 Pseudo-Spectral Discretization

WecOptTool uses a pseudo-spectral method that represents the WEC state $x(t)$ via a truncated Fourier series. Let f_1 be the fundamental frequency and N_f the number of harmonics. The state vector is

$$\boldsymbol{x}_{\text{wec}} = \big[\text{Re}(\hat{X}_1),\ \text{Im}(\hat{X}_1),\ \ldots,\ \text{Re}(\hat{X}_{N_f})\big]^\top \in \mathbb{R}^{2N_f-1}, \tag{2}$$

where the imaginary part of the highest frequency is excluded (Nyquist constraint). Derivatives and time-domain evaluations are computed via the time matrix $\boldsymbol{T}$ and derivative matrix $\boldsymbol{D}$:

$$x(t_k) = \boldsymbol{T}\,\boldsymbol{x}_{\text{wec}}, \qquad \dot{x}(t_k) = \boldsymbol{T}\boldsymbol{D}\,\boldsymbol{x}_{\text{wec}}. \tag{3}$$

The BEM-dependent forces in the time domain are:

$$\boldsymbol{f}_{\text{rad}} = \boldsymbol{T}\,\boldsymbol{Z}_{\text{rad}}\,\boldsymbol{x}_{\text{wec}}, \qquad \boldsymbol{Z}_{\text{rad}} = \text{blkdiag}\big(j\omega_k B(\omega_k) - \omega_k^2 A(\omega_k)\big), \tag{4}$$

$$\boldsymbol{f}_{\text{hs}} = \boldsymbol{T}\,\boldsymbol{Z}_{\text{hs}}\,\boldsymbol{x}_{\text{wec}}, \qquad \boldsymbol{Z}_{\text{hs}} = \text{blkdiag}(K), \tag{5}$$

$$\boldsymbol{f}_{\text{fric}} = \boldsymbol{T}\,\boldsymbol{Z}_{\text{fric}}\,\boldsymbol{x}_{\text{wec}}, \qquad \boldsymbol{Z}_{\text{fric}} = \text{blkdiag}(j\omega_k B_f), \tag{6}$$

$$\boldsymbol{f}_{\text{exc}} = \boldsymbol{T}\,\text{c2r}\Big(\textstyle\sum_\theta \hat{\eta}(\omega,\theta)\,X_{\text{exc}}(\omega,\theta)\Big), \tag{7}$$

where blkdiag$(\cdot)$ denotes the real block-diagonal representation of complex MIMO transfer matrices (mapping Fourier coefficients to Fourier coefficients), and c2r$(\cdot)$ converts complex Fourier coefficients to the real state representation.

2.3 Dynamics Residual

The dynamics equality constraint is the residual

$$r(\boldsymbol{x}_{\text{wec}}, \boldsymbol{x}_{\text{opt}}; \boldsymbol{h}) = \underbrace{-\omega^2 M\,\boldsymbol{x}_{\text{wec}}}_{\text{inertia}} - \underbrace{\boldsymbol{f}_{\text{rad}}(\boldsymbol{h})}_{\text{radiation}} - \underbrace{\boldsymbol{f}_{\text{hs}}(\boldsymbol{h})}_{\text{hydrostatics}} - \underbrace{\boldsymbol{f}_{\text{fric}}(\boldsymbol{h})}_{\text{friction}} - \underbrace{\boldsymbol{f}_{\text{exc}}(\boldsymbol{h})}_{\text{excitation}} - \underbrace{\boldsymbol{f}_{\text{add}}(\boldsymbol{x}_{\text{wec}}, \boldsymbol{x}_{\text{opt}})}_{\text{PTO, buoyancy, }\ldots} = \boldsymbol{0}, \tag{8}$$

where $\boldsymbol{h} = (A, B, K, B_f, F_{\text{FK}}, F_{\text{diff}}, M)$ collects all BEM-dependent hydrodynamic parameters, and $\boldsymbol{f}_{\text{add}}$ includes forces that do *not* depend on $\boldsymbol{h}$ (PTO, buoyancy, gravity, mooring).

3 Bilevel Optimization Problem

3.1 Inner Problem: Optimal WEC Control

For a fixed hull shape (and hence fixed BEM coefficients $\boldsymbol{h}$), the optimal control problem is the NLP:

$$\boxed{\begin{aligned} \varphi^*(\boldsymbol{h}) = \min_{\boldsymbol{x}_{\text{wec}}, \boldsymbol{x}_{\text{opt}}} \quad & f(\boldsymbol{x}_{\text{wec}}, \boldsymbol{x}_{\text{opt}}) \\ \text{subject to} \quad & r(\boldsymbol{x}_{\text{wec}}, \boldsymbol{x}_{\text{opt}}; \boldsymbol{h}) = \boldsymbol{0} && \text{(dynamics, } \boldsymbol{\lambda}) \\ & g_i(\boldsymbol{x}_{\text{wec}}, \boldsymbol{x}_{\text{opt}}) \geq \boldsymbol{0}, \quad i = 1, \ldots, m && \text{(inequality constraints, } \boldsymbol{\mu}_i) \end{aligned}} \tag{9}$$

where f is the objective (e.g., average electrical power from the PTO), $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}_i$ are the Lagrange multipliers returned by IPOPT.

For the AquaHarmonics device, the inequality constraints include:

1. Peak PTO torque limit: $\tau_{\max} - |\tau(t)| \geq 0$
2. Maximum rotational speed: $\dot{\phi}_{\max} - |\dot{\phi}(t)| \geq 0$
3. Maximum mechanical power: $P_{\max} - |P_{\text{mech}}(t)| \geq 0$
4. Minimum mooring line tension: $T(t) + T_{\min} \geq 0$
5. Zero mean heave position: $\bar{x} = 0$ (equality)

3.2 Outer Problem: Hull Shape Optimization

The outer problem adjusts the hull shape parameters $\boldsymbol{\theta} \in \mathbb{R}^{n_\theta}$ to maximize the optimal power:

$$\max_{\boldsymbol{\theta}} \quad \varphi^*\big(\boldsymbol{h}(\boldsymbol{\theta})\big) \qquad \text{subject to} \qquad \boldsymbol{\theta}^{\min} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}^{\max}. \tag{10}$$

For the AquaHarmonics device, we parameterize the hull by:

- $\theta_1 = r_1$: radius of the main (upper) cylinder [m], $r_1 \in [0.5, 2.5]$
- $\theta_2 = T_1$: draft of the main cylinder [m], $T_1 \in [0.5, 3.0]$

with default values $r_1 = 1.085\,\mathrm{m}$ and $T_1 = 1.5\,\mathrm{m}$. The remaining dimensions (T_2, T_3, r_2, r_3) are held fixed.

4 Gradient Computation

The total derivative of optimal power with respect to shape is

$$\frac{d\varphi^*}{d\theta_j} = \sum_i \frac{d\varphi^*}{dh_i} \cdot \frac{dh_i}{d\theta_j}, \qquad j = 1, \ldots, n_\theta. \tag{11}$$

We compute the two factors using different strategies.

4.1 Analytical: Fiacco Post-Optimality Sensitivity

The Fiacco sensitivity theorem [Fiacco, 1976] (also known as the envelope theorem) states that at an optimal solution $(\boldsymbol{x}^*_{\mathrm{wec}}, \boldsymbol{x}^*_{\mathrm{opt}})$ with Lagrange multipliers $\boldsymbol{\lambda}^*$, $\boldsymbol{\mu}^*_i$, the total derivative of the optimal objective with respect to a parameter h is:

$$\frac{d\varphi^*}{dh} = \left.\frac{\partial f}{\partial h}\right|_{x^*} + \boldsymbol{\lambda}^{*\top} \left.\frac{\partial r}{\partial h}\right|_{x^*} + \sum_{i=1}^{m} \boldsymbol{\mu}_i^{*\top} \left.\frac{\partial g_i}{\partial h}\right|_{x^*}. \tag{12}$$

For the AquaHarmonics device, several simplifications apply:

1. The objective $f = \mathrm{pto.average_power}$ depends only on $(\boldsymbol{x}_{\mathrm{wec}}, \boldsymbol{x}_{\mathrm{opt}})$ and not on $\boldsymbol{h}$ directly, so $\frac{\partial f}{\partial \boldsymbol{h}} = \boldsymbol{0}$.
2. The inequality constraints (torque, speed, power, tension) depend only on $(\boldsymbol{x}_{\mathrm{wec}}, \boldsymbol{x}_{\mathrm{opt}})$ and not on $\boldsymbol{h}$, so $\frac{\partial g_i}{\partial \boldsymbol{h}} = \boldsymbol{0}$ for all i.

Therefore, the Fiacco formula reduces to:

$$\boxed{\frac{d\varphi^*}{d\boldsymbol{h}} = \boldsymbol{\lambda}^{*\top} \left.\frac{\partial r}{\partial \boldsymbol{h}}\right|_{(\boldsymbol{x}^*_{\mathrm{wec}}, \boldsymbol{x}^*_{\mathrm{opt}})}} \tag{13}$$

Implementation via VJP. The vector-Jacobian product (VJP) $\boldsymbol{\lambda}^{*\top} \frac{\partial r}{\partial \boldsymbol{h}}$ is computed efficiently using reverse-mode automatic differentiation (JAX). Define

$$r_h(\boldsymbol{h}) \;=\; r(\boldsymbol{x}^*_{\mathrm{wec}}, \boldsymbol{x}^*_{\mathrm{opt}}; \boldsymbol{h}), \tag{14}$$

where the WEC state is held fixed at the optimum. The VJP is

$$\frac{d\varphi^*}{d\boldsymbol{h}} = \mathrm{vjp}(r_h, \boldsymbol{h})^\top \boldsymbol{\lambda}^*. \tag{15}$$

The function r_h is implemented as `residual_parametric`—a pure JAX function that recomputes the BEM-dependent forces (radiation, hydrostatics, friction, excitation) from the parameter arrays, without calling the original force closures. This makes $\partial r / \partial \boldsymbol{h}$ available to JAX's AD system.

Computational cost. A single VJP evaluation costs $\mathcal{O}(1)$ times the cost of evaluating r itself, regardless of the number of BEM parameters n_h. For the AquaHarmonics device with $n_h = 63$ real-valued entries, this is essentially instantaneous compared to the IPOPT solve.

4.2 Finite Differences: BEM Jacobian

Since Capytaine (the BEM solver) is not differentiable, we estimate $\partial \boldsymbol{h}/\partial \theta_j$ via central finite differences:

$$\frac{dh_i}{d\theta_j} \approx \frac{h_i(\boldsymbol{\theta} + \epsilon\, \boldsymbol{e}_j) - h_i(\boldsymbol{\theta} - \epsilon\, \boldsymbol{e}_j)}{2\epsilon}, \tag{16}$$

where $\epsilon = 10^{-3} \max(|\theta_j|, 0.1)$ and $\boldsymbol{e}_j$ is the j-th standard basis vector.

Computational cost. Each shape parameter requires two BEM evaluations (forward and backward perturbation). For $n_\theta = 2$ shape parameters, this is 4 BEM solves per gradient evaluation. Crucially, these BEM solves do *not* require re-solving the IPOPT problem—only the hydrodynamic coefficients are needed.

4.3 Hybrid Gradient via Chain Rule

Combining eqs. (13) and (16) into the chain rule (11):

$$\boxed{\frac{d\varphi^*}{d\theta_j} = \sum_{i=1}^{n_h} \underbrace{\frac{d\varphi^*}{dh_i}}_{\text{Fiacco (analytical)}} \cdot \underbrace{\frac{dh_i}{d\theta_j}}_{\text{FD (Capytaine)}}} \tag{17}$$

In practice, this inner product is computed as:

$$\frac{d\varphi^*}{d\theta_j} = \sum_{k \in \mathcal{P}} \mathrm{Re}\left\langle \frac{d\varphi^*}{d\boldsymbol{h}_k}, \frac{d\boldsymbol{h}_k}{d\theta_j} \right\rangle, \tag{18}$$

where $\mathcal{P} = \{A, B, K, B_f, F_{\mathrm{FK}}, F_{\mathrm{diff}}, M\}$ and the inner product accounts for complex parameters (Froude–Krylov and diffraction forces) via conjugation.

4.4 Cost Comparison

Table 1 compares the per-iteration cost of the hybrid gradient with a naive finite-difference approach that re-solves the full NLP for each shape perturbation.

Table 1: Computational cost per gradient evaluation (n_θ shape parameters).

Component	**Hybrid (ours)**	**Full FD**
IPOPT solves	1	$2n_\theta + 1$
BEM solves	$2n_\theta + 1$	$2n_\theta + 1$
VJP evaluations	1	0
Dominant cost	**1 IPOPT**	$\mathbf{2n_\theta + 1}$ **IPOPT**

For the AquaHarmonics device with $n_\theta = 2$, the hybrid method requires 1 IPOPT solve versus 5 IPOPT solves for full FD—a 5× reduction in the most expensive component. The advantage grows linearly with n_θ.

5 Optimization Algorithm

The complete shape optimization procedure is given in Algorithm 1.

Key implementation details:

Algorithm 1 Gradient-based hull shape optimization

Require: Initial shape $\boldsymbol{\theta}^{(0)}$, bounds $[\boldsymbol{\theta}^{\min}, \boldsymbol{\theta}^{\max}]$, FD step ϵ
Ensure: Optimized shape $\boldsymbol{\theta}^*$

$\boldsymbol{x}_{\text{warm}}^{(0)} \leftarrow \texttt{None}$
for $k = 0, 1, 2, \ldots$ until convergence **do**
  — **Shape-to-Power Pipeline** —
  $\boldsymbol{h}^{(k)} \leftarrow \texttt{Capytaine}(\boldsymbol{\theta}^{(k)})$ ▷ Mesh + BEM
  $(\boldsymbol{x}_{\text{wec}}^*, \boldsymbol{x}_{\text{opt}}^*, \boldsymbol{\lambda}^*) \leftarrow \texttt{IPOPT}(\boldsymbol{h}^{(k)}, \boldsymbol{x}_{\text{warm}}^{(k)})$ ▷ Inner NLP solve
  $\varphi^{(k)} \leftarrow f(\boldsymbol{x}_{\text{wec}}^*, \boldsymbol{x}_{\text{opt}}^*)$
  $\boldsymbol{x}_{\text{warm}}^{(k+1)} \leftarrow (\boldsymbol{x}_{\text{wec}}^*, \boldsymbol{x}_{\text{opt}}^*)$ ▷ Warm-start for next iteration
  — **Hybrid Gradient** —
  $\frac{d\varphi^*}{d\boldsymbol{h}} \leftarrow \boldsymbol{\lambda}^{*\top} \frac{\partial r}{\partial \boldsymbol{h}}\big|_{x^*}$ ▷ Fiacco VJP (analytical)
  for $j = 1, \ldots, n_\theta$ **do**
    $\boldsymbol{h}^+ \leftarrow \texttt{Capytaine}(\boldsymbol{\theta}^{(k)} + \epsilon\, \boldsymbol{e}_j)$
    $\boldsymbol{h}^- \leftarrow \texttt{Capytaine}(\boldsymbol{\theta}^{(k)} - \epsilon\, \boldsymbol{e}_j)$
    $\frac{d\varphi^*}{d\theta_j} \leftarrow \sum_i \frac{d\varphi^*}{dh_i} \cdot \frac{h_i^+ - h_i^-}{2\epsilon}$ ▷ Chain rule
  end for
  — **Outer Update** —
  $\boldsymbol{\theta}^{(k+1)} \leftarrow \texttt{L-BFGS-B}\big(\boldsymbol{\theta}^{(k)},\ -\nabla_\theta \varphi^*,\ [\boldsymbol{\theta}^{\min}, \boldsymbol{\theta}^{\max}]\big)$
end for

- **Warm-starting**: the IPOPT solver is warm-started from the previous iteration's solution, greatly reducing convergence time as shape changes are typically small between iterations.
- **Sign convention**: we minimize $-\varphi^*$ (negative power) since `scipy.optimize.minimize` performs minimization.
- **Bound constraints**: the L-BFGS-B algorithm natively handles the box constraints on $\boldsymbol{\theta}$.

6 Application: AquaHarmonics Device

6.1 Device Description

The AquaHarmonics WEC is a heave-mode point absorber with a complex drive train including:

- A multi-stage gear system (gear ratios R_{21}, R_{45}, R_{67})
- A pneumatic air spring
- A generator with Joule heating losses: $P_{\text{loss}} = R_w (T_e/k_T)^2$ where $R_w = 0.4\ \Omega$ and $k_T = 1.5$ Nm/A

The hull geometry consists of an upper cylinder (radius r_1, draft T_1), a conical frustum transition (height T_2), and a lower tube (height T_3, outer radius r_2, inner radius r_3).

6.2 Problem Setup

- **Wave conditions**: Regular wave, $A = 0.5\,\text{m}$, $f = 0.12\,\text{Hz}$, $\phi = 30°$
- **Frequency resolution**: $f_1 = 0.12\,\text{Hz}$, $N_f = 10$ harmonics
- **Shape parameters**: $\boldsymbol{\theta} = (r_1, T_1)$
- **Fixed geometry**: $T_2 = 0.355\,\text{m}$, $T_3 = 7.25\,\text{m}$, $r_2 = 0.405\,\text{m}$, $r_3 = 0.355\,\text{m}$

- **Mass**: $m = 5000\,\mathrm{kg}$ (fixed)
- **Optimizer**: L-BFGS-B via `scipy.optimize.minimize`

6.3 Baseline Sensitivity

Figure 1 shows the Fiacco sensitivity $\max_i |d\varphi^*/dh_i|$ for each BEM parameter group at the default hull shape. This reveals which hydrodynamic coefficients the optimal electrical power is most sensitive to, and motivates the choice of shape parameters that most affect these coefficients.

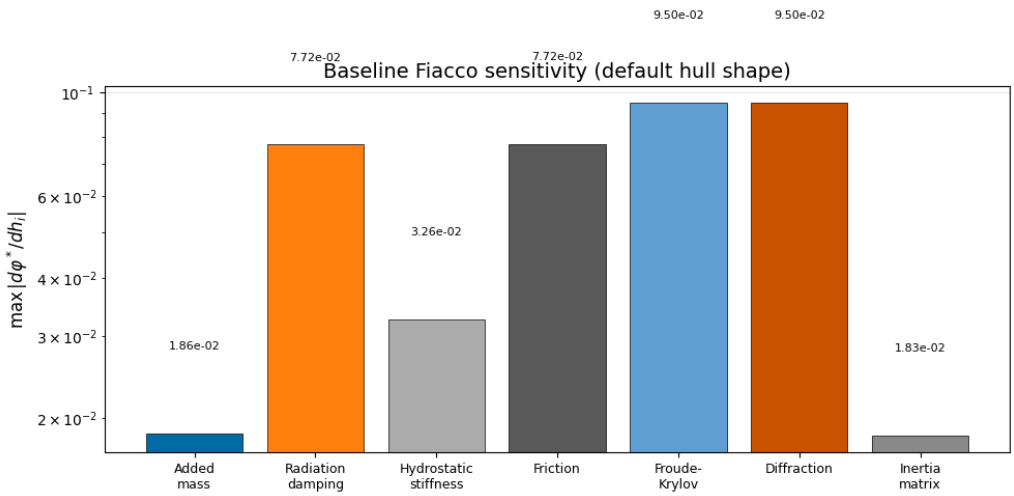

Figure 1: Fiacco sensitivity of optimal electrical power to each BEM parameter group at the baseline hull shape ($r_1 = 1.085\,\mathrm{m}$, $T_1 = 1.5\,\mathrm{m}$).

6.4 Optimization Results

The L-BFGS-B optimizer converges in a modest number of iterations. Figure 2 shows the evolution of optimal power and the two shape parameters across iterations.

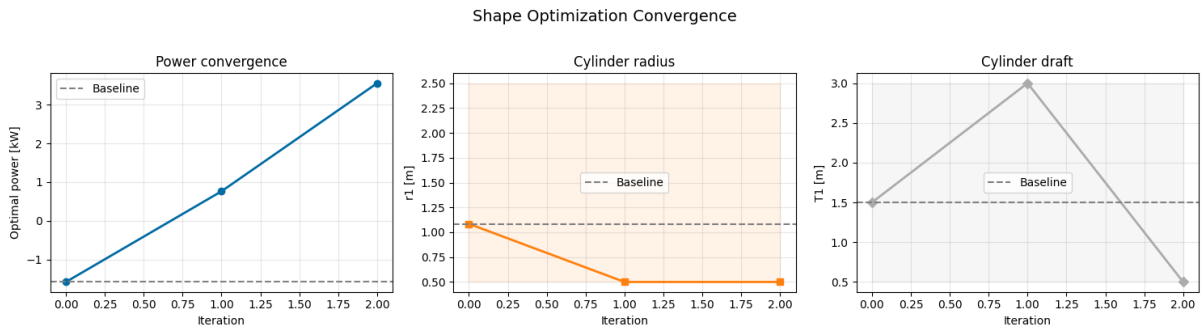

Figure 2: Convergence history of the shape optimization. Left: optimal electrical power vs. iteration. Center: cylinder radius r_1. Right: cylinder draft T_1. Dashed lines indicate baseline values; shaded regions indicate the feasible bounds.

Figure 3 compares the hull cross-sections of the baseline and optimized designs.

Figure 4 compares the Fiacco sensitivity at the baseline and optimized shapes, revealing how the sensitivity profile changes with hull geometry.

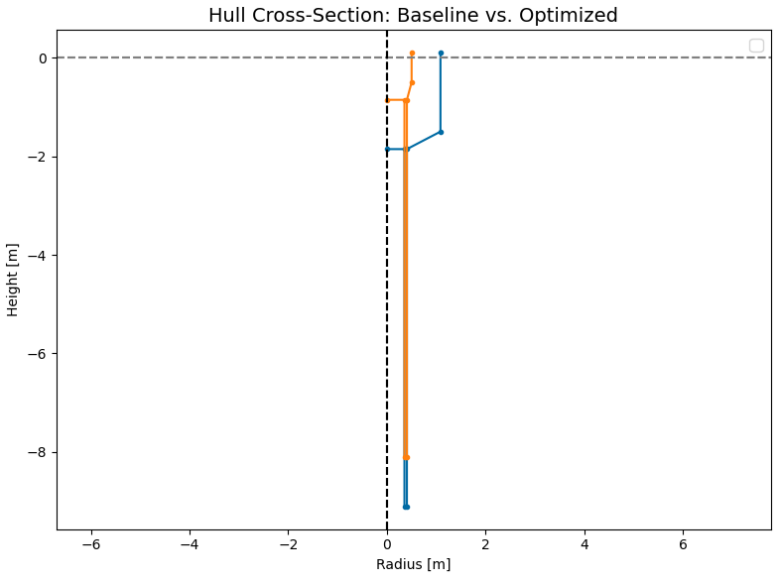

Figure 3: Hull cross-section comparison: baseline (blue) vs. optimized (orange) AquaHarmonics geometry.

7 Software Implementation

The framework is implemented as extensions to WecOptTool [Bacelli et al., 2022], an open-source Python toolbox for WEC design optimization. The key modules are:

- `wecopttool.solver_ipopt`: Provides `WEC_IPOPT`, a subclass of `WEC` that uses IPOPT (via `cyipopt`) and returns Lagrange multipliers. Also provides `sensitivity()` for the Fiacco VJP and `make_differentiable_solver()` for a JAX-native interface with `jax.custom_vjp`.
- `wecopttool.parametric`: Provides `residual_parametric`—a pure JAX re-implementation of the BEM-dependent dynamics residual, compatible with `jax.vjp` and `jax.grad`.
- `wecopttool.sensitivity_plots`: Reusable visualization utilities for sensitivity bar charts, per-frequency sensitivity plots, and analytical vs. finite-difference comparison plots.

Dependencies include JAX (automatic differentiation), Capytaine (BEM solver), IPOPT/cyipopt (NLP solver), and pygmsh/gmsh (mesh generation).

8 Discussion

8.1 Advantages

1. **Computational efficiency**: The hybrid gradient eliminates the need to re-solve the NLP for each finite difference perturbation. The IPOPT solve (the dominant cost) is performed only once per outer iteration.
2. **Gradient accuracy**: The Fiacco sensitivity is exact (to machine precision for the constraint term), while only the BEM Jacobian is approximated via finite differences.
3. **Scalability**: The cost of the Fiacco VJP is independent of the number of BEM parameters ($n_h \approx 63$ for the AquaHarmonics device). The FD cost scales with n_θ (shape parameters), not n_h.
4. **Black-box BEM**: Capytaine is treated as a black box—no source-code modification or adjoint implementation is required.

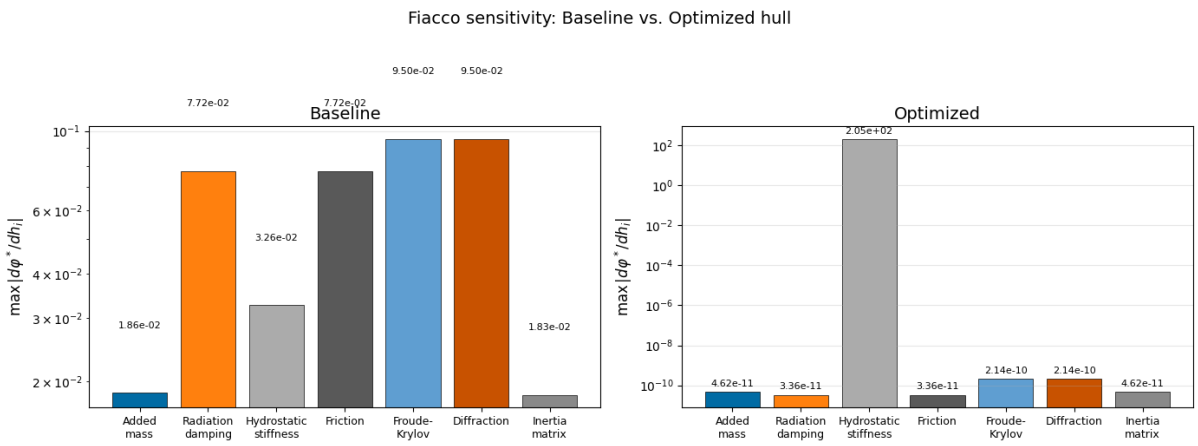

Figure 4: Fiacco sensitivity comparison between baseline and optimized hull shapes. The sensitivity profile shifts as the optimizer adjusts the hull to maximize power.

8.2 Limitations and Future Work

1. **BEM finite differences**: The FD approximation of $\partial \boldsymbol{h}/\partial \boldsymbol{\theta}$ introduces truncation error and requires careful step-size selection. Differentiable BEM solvers (e.g., via AD through Capytaine or an adjoint method) would eliminate this entirely.

2. **Single wave condition**: The current demonstration uses a single regular wave. Extension to irregular sea states and multiple realizations is supported by the multi-realization feature in `make_differentiable_solver`.

3. **Local optima**: The L-BFGS-B algorithm finds local optima. Multi-start strategies or global optimization methods could be combined with the gradient information.

4. **Additional shape parameters**: The current demonstration optimizes only r_1 and T_1. The framework naturally extends to more parameters (T_2, T_3, r_2, r_3, or even free-form deformations).

9 Conclusion

We have presented a hybrid analytical–finite-difference framework for gradient-based hull shape optimization of wave energy converters. By exploiting the Fiacco post-optimality sensitivity theorem, the method avoids the dominant cost of re-solving the inner NLP for each shape perturbation. The framework is implemented as an extension to WecOptTool and demonstrated on the AquaHarmonics device (fig. 3), where the optimizer converges in a modest number of iterations (fig. 2) to an improved hull shape.

The key insight is that the bilevel structure of WEC design—hull shape determines hydrodynamic coefficients, which determine optimal control and power—can be efficiently exploited by computing exact sensitivities through the inner optimization and approximating only the non-differentiable BEM component via finite differences.

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